Advanced Labs I　(AY2025)

Theme			
B. Electrical Conductivity and Hall Effect of Semiconductors			
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Date of Receipt	
Recipient	

By the theme instructor,

Date of return	
Rating /10	
Are the principles, equipment, and experimental procedures fully understood? a b c d e	
Are the experimental data properly analyzed? a b c d e	
Are the experimental results presented appropriately? a b c d e	
Is the discussion substantial? a b c d e	
Is the structure of objectives, methods, results, discussion, etc. appropriate? a b c d e	
Is it carefully described? a b c d e	
Comments	

Advanced Labs 1 Theme B. Electrical Conductivity and Hall Effect of Semiconductors

1 Experiment 1 (Day 1)

1.1 Purpose On Day 1, Part one focuses on using the four-terminal method to carry out measurements and confirm Ohm's law. Part two helps us learn the hall effect and calculate the Hall coefficient, free carrier concentration, and mobility by determining whether the carrier is an electron or hole.

Part 1: Four terminal method and resistivity calculations (Current-Voltage Characteristics)

1.2 Methods In this experiment, DC current was flowed through a sample and the voltage (Vs) across both ends of a standard resistor (Rs) was measured using a digital voltmeter.

1.3 Results and Discussion The Voltage was recorded to generate a V-I graph as shown in Figure 1 in the range of −30mA < Ix < 30mA.

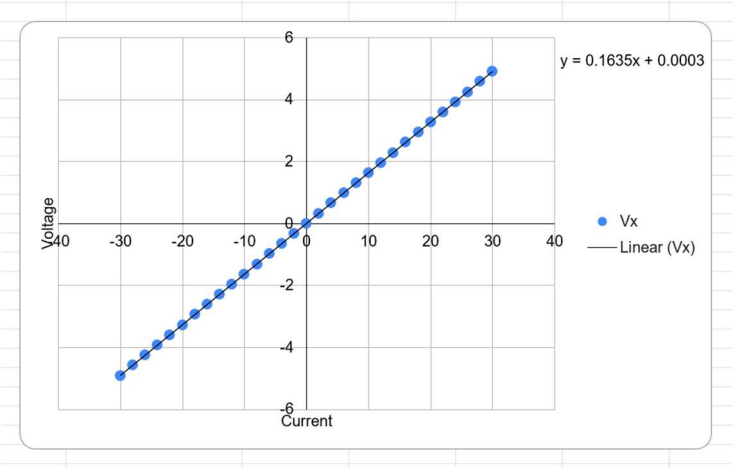


Figure 1 Voltage [V] vs Current [Ix]

The resistance was derived from the slope using Ohm's law principle,

Vs[V] = Is[A] · Rs[Ω]Rs = 0.1635[Ω].

The resistivity (ρ) can be derived using the following formula, ρ[Ω · m] = R[Ω]·A[m2]/ l[m] = 0.1635· (0.0015× 0.001)/ 0.003=0.00008175≈ 8.18×10−5[Ω·m]

where R is resistance, A is the cross-sectional area and l is the length of the sample. Finally, the conductivity (σ) can be calculated from the following relation,

σ[Ω−1 · m−1] = 1/ ρ[Ω · m] = 1/0.00008175= 12232.4159≈ 1.22×10^4[Ω−1 ·m−1]

A clear Ohm's law relationship was established, suggesting that there was no significant systemic error in the

Voltage and Current values at a constant room temperature and we were familiarized with the equipment.

Part 2: Hall Effect experiment (Magnetic field dependence of Hall Voltage)

1.4 Methods

In the second part, the sample was connected to a holder attached to a cryostat and the magnetic field dependence of the Hall Voltage was measured by setting the current Ix to +15mA and the magnetic field B to +0.2T
and the Hall voltage is measured as the field is reduced to-0.2T. This procedure is repeated for B = −0.2T and increased gradually to zero. Finally the above steps are repeated for a current of-15mA.

1.5 Results and Discussion

The results for the-15mA current are plotted in Figure 2. 15mA results are similar (absolute values of slope for the -15 mA and 15mA are close in value) but with a negative slope of 3.681. (Figure 3) : (Hall voltage : vertical axis, magnetic field, horizontal axis)

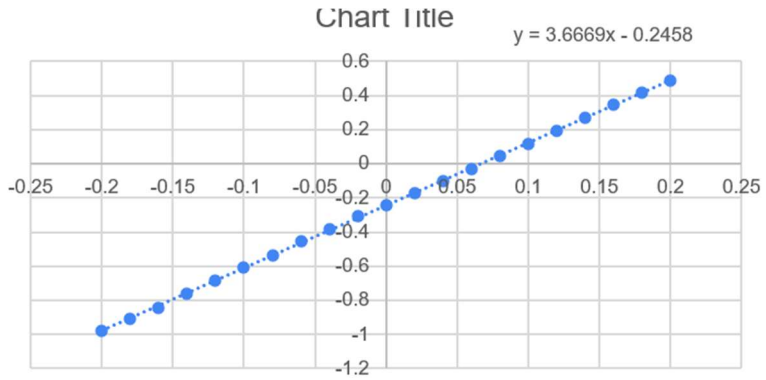


Figure 2 Hall voltage [V] versus Magnetic Field [T] for -15mA

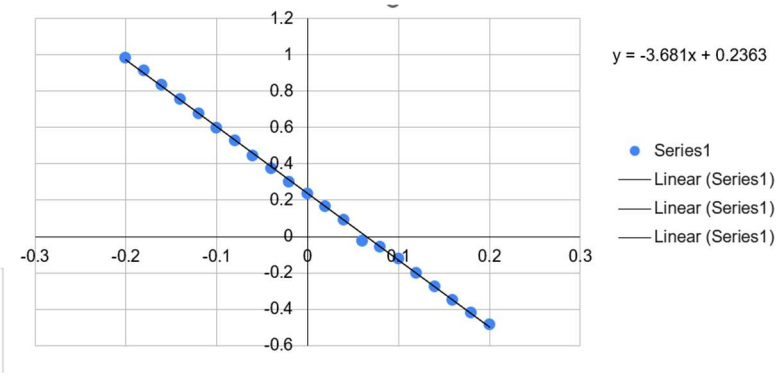


Figure 3 Hall voltage [V] versus Magnetic Field [T] for 15mA

The Hall coefficient is calculated as the following:

For 15mA,

Rh (+)

$$= \frac{E_y}{J_x \cdot B_z} = \frac{b[m] \cdot V_y[V]}{I_x[A] \cdot B_z[T]} \quad (1)$$

= (0.001 x -3.681)/ 0.0015 = -0.0002454 =−2.45×10−4[m^3/C]

For -15 mA,

Rh (-)

$$= \frac{E_y}{J_x \cdot B_z} = \frac{b[m] \cdot V_y[V]}{I_x[A] \cdot B_z[T]}$$

= (0.001 x -3.6669)/ -0.0015 = -0.00024446=−2.45×10−4[m^3/C]

where Vy/Bz is the V-B slope found in Figures 2 and 3.

Since the Hall coefficient was found to be negative, it can be determined that the carrier was an electron. Thus, it is an n-type semiconductor. From the equation below, the carrier concentration was found to be

$$n = \frac{1}{R_H q} \qquad (2)$$

n= 1/(-0.000245 x1.602x10^-19) ≈2.55 x 10^22

Using the conductivity found in Part 1 and the Hall coefficient, the mobility was calculated to be,

$$\mu = |R_H| \cdot \sigma$$

= |−0.000245| x (1.22×10^4) = 2.989 ≈ 3.00 [m^2/V ·s]

The value for the mobility calculated through Excel is 2.996941896, which is close to the value calculated through the equation.

Finally, using the information derived above and the values m/m0 = 0.014, m0 = 9.11 × 10^−31,|q| = e = 1.6 ×10^−19, the carrier relaxation time can also be calculated.

$$\tau = \frac{\mu \cdot m}{|q|} = \frac{\mu \cdot m \cdot (m/m_0)}{|e|}$$

= (2.996941896·0.014 · 9.11 × 10^−31)/(|1.6 × 10−19| = 2.39 ×10^-13s There was a clear linear relationship established between Magnetic Field and Hall Voltage which matches with the theoretically established relationship between the two variables. The carrier was found to be an electron with a relaxation time of 2.39 × 10−13s.

2 Experiment 2 (Day 2)

2.1 Purpose

The purpose of the Day 2 experiment was to measure the temperature dependency of resistivity and to measure the energy gap EG (forbidden band energy range of semiconductors) and study its impact on the electrical conductivity of the semiconductor.

2.2 Methods

On Day 2, current (Ix) was flown into the sample, the voltage (Vx) was measured and the resistivity was calcu lated from Vx = [Vx(+)−Vx(−)]/2 for temperatures ranging from 293K to 103K. Values at lower temperatures were measured using a cryostat to cool the sample using liquid nitrogen.

2.3 Results and Discussion

A Table was created with the values of Vx and resistivity ρ and the values were plotted in a resistivity vs temperature graph shown in Figure 4.

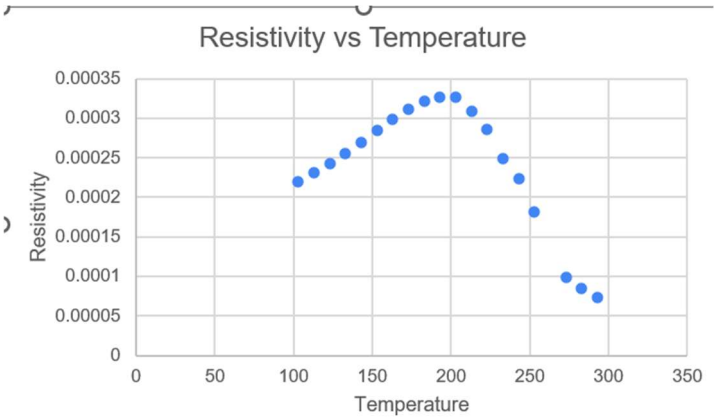


Figure 4 Resistivity vs Temperature

Considering Figure 3-4 in the textbook, the plotted graph can be considered most similar to the intrinsic region C-B. ln(ρ) vs 1000/T was plotted on a graph (Figure 5) to analyze the linearized results equivalent to a semi-log graph.

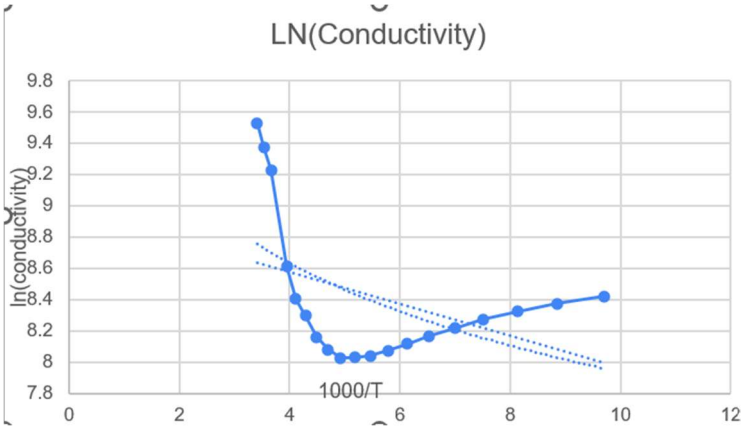


Figure 5 Semilogarithmic graph of resistivity in relation to 1000/T [K−1]

Figure 6 shows the region from 203K to 293K as with a linear fit

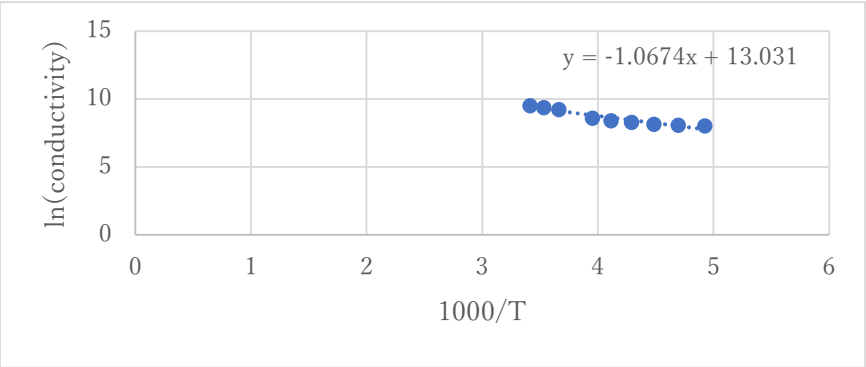


Figure 6 Semilogarithmic graph of resistivity with 1000/T [K−1] (linear fit)

Considering Equation (19) in the textbook,

$$\sigma \propto exp\left(-\frac{E_G}{2kT}\right)$$

Since $\sigma = 1/\rho$, we get,

$$\rho \propto exp\left(\frac{E_G}{2kT}\right)$$

$$ln(\rho) = \frac{E_G}{2kT} + ln(K) \text{where K is a constant}$$

$$ln(\rho) = \frac{E_G}{2k} \times \frac{1}{T} + ln(K)$$

Comparing the slope of the graph in Figure 6 with the equation above, we can see that the slope is EG/2k(1000).

From this, we can derive the energy gap EG [eV].

$\frac{E_G}{2000k}$= (-1.0674) (-2000)(0.00008617087)

$E_G = -1.0674 \cdot -2000 \cdot 0.00008617087 = 0.18$ eV

where k is the Boltzmann Constant (k = 8.6173×10^−5[eV])

Comparing these with known values, this value is closest to the energy gap of Indium Antimonide (InSb) ≈ 17 eV. Although the measured value is higher, it could be due to experimental errors such as approximations made of temperature readings and the methodology of cooling the material from high temperature to low temperature rather than the other way around.

3 Experiment 3 (Day 3)

3.1 Purpose

The purpose of Day 3 was to measure the temperature dependency of the Hall coefficient and derive the carrier concentration and mobility temperature dependency. We also looked at the mechanism of electrical conductivity in semiconductors and measured the magnetic field dependency of Hall coefficient at low temperatures.

Part 1: Measuring the temperature dependency of the Hall coefficient

3.2 Methods In part one, the current was set to-15mA and the temperature was changed from 293K to

103K using a cryostat and liquid nitrogen. The Voltage was measured with the configurations VEF(+Ix,+B),VEF(+Ix,−B), VEF(−Ix,+B),VEF(−Ix,−B) and the real Hall Voltage (Vy) was calculated using the following equation, Vy = 1 4VEF(+Ix,+B)−VEF(+Ix,−B)−VEF(−Ix,+B)+VEF(−Ix,−B) where VEF is the measured voltage, Ix is current and B is the magnetic field.

3.3 Results and Discussion

The Hall coefficients for each temperature were calculated using Vy and Equation 1 and was plotted against temperature to get the graph shown in Figure 7.

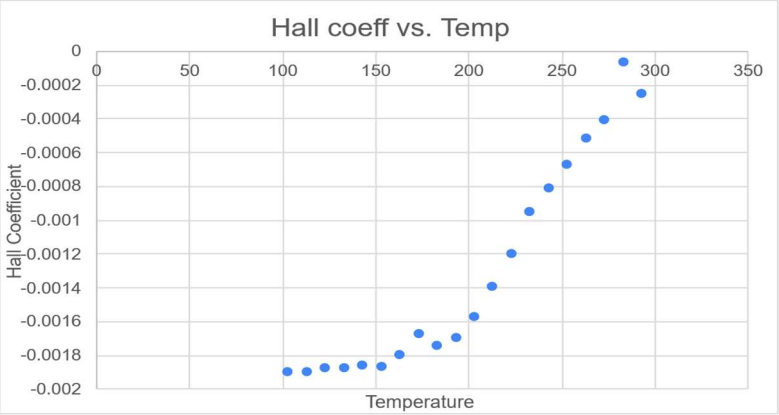


Figure 7 Hall Coefficient [mV/AT] versus Temperature [K]

The carrier concentrations were calculated using Equation 2 where q = e =9.11×10^−19.

Following this, ln(RH and n were plotted against 1000/T equivalent to a linear semi-log graph in Figures 8, 9 respectively.

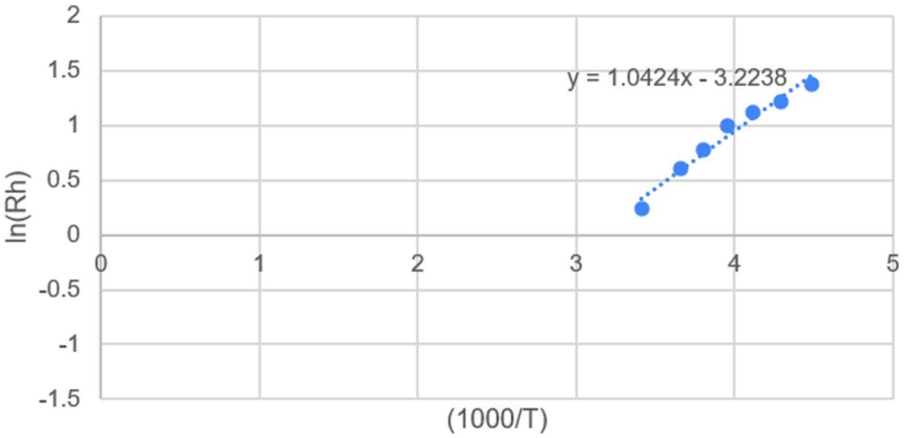


Figure 8 Semilogarithmic relation of Hall Coefficient RH[mV/AT] and Temperature [K−1]

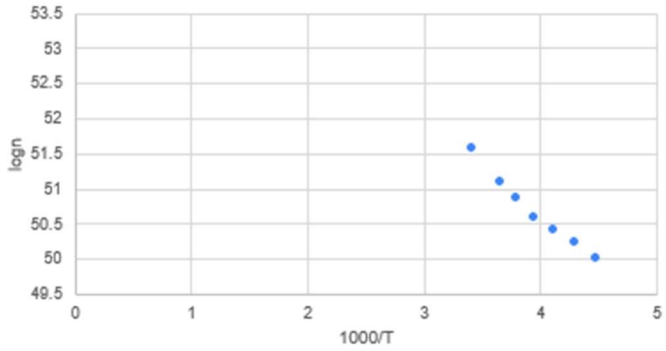


Figure 9 Semilogarithmic relation of carrier concentration (n) and 1000/T [K−1]

Considering the Figure 3-4 in the textbook, the obtained curves posses similarity to intrinsic region C-B. The energy gap EG was derived using a similar method as Day 2. Using Equation (18) from the textbook,

$$R_H \propto exp\left(-\frac{E_G}{2kT}\right)$$

Therefore, the gradient of the RH versus 1000/T graph is given by EG/2k · 1000.

$$\frac{E_G}{2000k} = 1.0424$$

$$E_G = 1.0424 * (2000 \cdot k) = 1.0424 \cdot 2000 \cdot 8.6173 \times 10\text{^} - 5 = 0.18eV$$

This result, when rounded, is the same as in Day 2.

Using the relation μ = RH/ρ, the mobility can be derived. Finally the log of mobility ln(μ) was plotted against the log of temperature ln(T) to derive the value of γ.

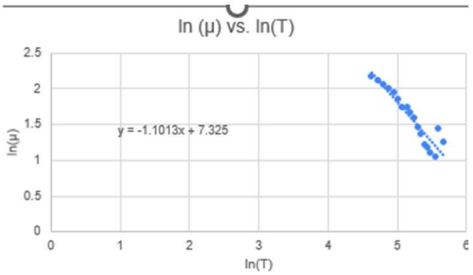


Figure 10 ln(μ) versus ln(T

Based on the relationship μ ∝ T^γ,

ln(μ) = γln(T) +ln(K) where K is a constant

From the above equation, it is established that the gradient of Figure 10 is equal to γ.

Thus, γ= -1.1013

According to Matthesen's law, as mobility is proportional to carrier relaxation time the reciprocal of total mobility is expressed as the sum of the mobility reciprocals due to the various scattering mechanisms.

$$\frac{1}{\mu} = \frac{1}{\mu_{lattice} + \frac{1}{\mu_{impurity}}}$$

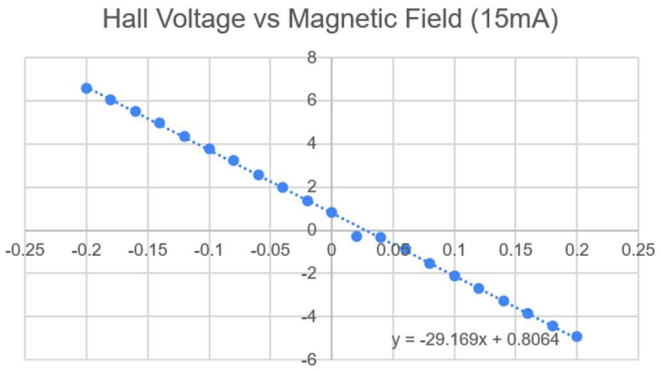
Since electrical conductivity is proportional to mobility, we get a similar relation for mobility and conductivity, suggesting that at low temperatures, conductivity is mostly constant because impurity scattering dominates. At higher temperatures (room temperature), lattice scattering dominates causing electrical conductivity to decrease. The value of γ characterizes how mobility and conductivity changes with temperature. A higher γ implies a stronger temperature dependence (mobility and conductivity decreases more rapidly with increasing temperature). Since the value of γ was found to be close to 1, it can be said that the given sample had a weak temperature dependence perhaps because of impurity scattering or other weak scattering mechanisms.

Part 2: Measuring Hall Coefficient at Low Temperature

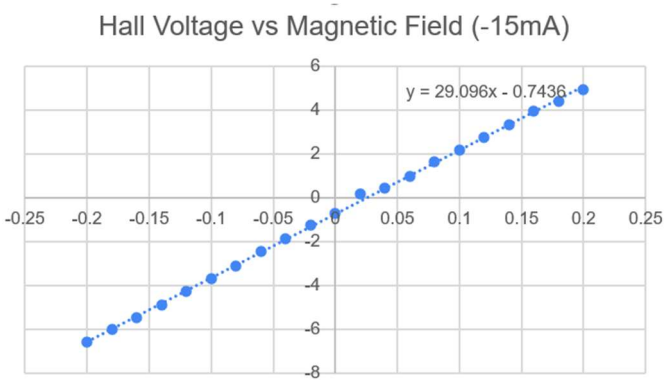
3.4 Methods In the second part, liquid nitrogen was injected into the cryostat and the magnetic field dependency of the Hall Voltage was measured at a low temperature ≈ 100K using the same process as Day 1, Part 1.

3.5 Results and Discussion

The graph of Hall Voltage [V EF] for 15 mA: (Hall voltage : vertical axis, magnetic field, horizontal axis)



The graph of Hall Voltage [V EF] for -15 mA



Using Equation (1),

The Hall coefficient for -15mA is

Rh(-)= (0.01 x 29.096)/-0.015 = -0.00194 [m^3/C]

Rh(+)=(0.01 x -29.169)/-15=-0.001945[m^3/C]

Since the coefficient is negative, the carrier is an electron and the semiconductor is n-type.

The carrier concentration, electrical conductivity, and mobility can be derived from:

$$n = \frac{1}{R_H q}$$

n=1/(0.00194 x 1.602 x 10^-19) = 3.22 x 10^21

To calculate electrical conductivity:

The resistivity is given from the Day 3 data where 15mA and 0T condition was used for the system.

$$\rho = 9.076 \times 10^{-5}\ \Omega \cdot \mathrm{m}$$

$$\sigma = \frac{1}{\rho}$$

$$= \frac{1}{9.076 \times 10^{-5}}$$

$$= 1.10181 \times 10^{4}\ \Omega^{-1}\,\mathrm{m}^{-1}$$

$$\mu = |R_H| \cdot \sigma$$

= (0.00194)(1.10181 x 10^-4) = 21.375114 [m^2V^-1s^-1]